

# ABRAR SAMIN

Computer Science & Engineering Graduate · AI/ML Researcher · Teaching Assistant

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## PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from BRAC University (CGPA 3.97/4.00), graduating among the top of the department, with four IEEE-published research papers in machine learning, computer vision, and explainable AI. Served as an Undergraduate Teaching Assistant for both Data Structures and Algorithms, mentoring 200+ students, and contributed to BRAC University-funded research under the Research Seed Grant Initiative (RSGI), including AI-based assistive technology for accessibility. Strong foundation in machine learning, NLP, computer vision, and full-stack development, with a keen interest in teaching, mentoring, and advancing applied AI research in a university setting.

## RESEARCH INTERESTS

Machine Learning · Computer Vision · Explainable AI (XAI) · Embedded / Edge AI · Assistive Technology · Biomedical Signal Processing

## EDUCATION

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| <b>BRAC University</b><br><i>B.Sc. in Computer Science and Engineering — Dhaka, Bangladesh</i><br>CGPA: 3.97 / 4.00 — graduated among the top of the department | 2022 – 2026 |
| <b>Government Science College</b><br><i>Higher Secondary Certificate (HSC) — Dhaka, Bangladesh</i><br>GPA: 5.00 / 5.00                                          | 2019 – 2021 |
| <b>Adamjee Cantonment College</b><br><i>Secondary School Certificate (SSC) — Dhaka, Bangladesh</i><br>GPA: 5.00 / 5.00                                          | 2017 – 2019 |

## PUBLICATIONS

### JOURNAL ARTICLES

[J1] Reshad UI Karim, Sammam Mahdi, **Abrar Samin**, Aniqua Nusrat Zereen, M. Abdullah-Al-Wadud, and Jia Uddin, “Optimizing Stroke Recognition with MediaPipe and Machine Learning: An Explainable AI Approach for Facial Landmark Analysis,” *IEEE Access*, vol. 13, pp. 32636–32660, 2025, doi: [10.1109/ACCESS.2025.3550577](https://doi.org/10.1109/ACCESS.2025.3550577).

### CONFERENCE PAPERS

[C1] Tasnim Ferdous, Reshad UI Karim, **Abrar Samin**, Sammam Mahdi, Himika Tasnim, and Aniqua Nusrat Zereen, “Machine Learning Approaches in Photoplethysmography-Based Sleep Stage Classification,” in *Proc. 2024 IEEE 2nd Int. Conf. on Electrical, Automation and Computer Engineering (ICEACE)*, Changchun, China, Dec. 2024, pp. 123–128, doi: [10.1109/ICEACE63551.2024.10898858](https://doi.org/10.1109/ICEACE63551.2024.10898858).

[C2] Tasnim Ferdous, Reshad UI Karim, **Abrar Samin**, Sammam Mahdi, and Aniqua Nusrat Zereen, “Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning,” in *Proc. 2024 IEEE 2nd Int. Conf. on Electrical, Automation and Computer Engineering (ICEACE)*, Changchun, China, Dec. 2024, pp. 117–122, doi: [10.1109/ICEACE63551.2024.10898853](https://doi.org/10.1109/ICEACE63551.2024.10898853).

[C3] Sammam Mahdi, Reshad UI Karim, Tareq Sujat, Syeda Maliha Tabassum, **Abrar Samin**, and Aniqua Nusrat Zereen, “Vision-based Hand Gesture Virtual Keyboard-Mouse framework with Bilingual Next-word prediction,” in *Proc. 2026 23rd Int. Joint Conf. on Computer Science and Software Engineering (JCSSE)*, Bangkok, Thailand, Jun. 2026, pp. 638–643, doi: [10.1109/JCSSE68839.2026.11596610](https://doi.org/10.1109/JCSSE68839.2026.11596610).

## RESEARCH EXPERIENCE

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| <b>Undergraduate Researcher</b><br><i>Research Seed Grant Initiative (RSGI), BRAC University — Dhaka, Bangladesh</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2024 – Present |
| <ul style="list-style-type: none"><li>Contributed to a BRAC University-funded research project under the Research Seed Grant Initiative (RSGI) developing an AI-powered assistive navigation system to support safe, independent mobility for blind and low-vision users.</li><li>Worked on computer vision and on-device (edge) AI components for a low-cost wearable platform, advancing the system from a validated proof-of-concept toward a deployable prototype intended for real-world, ethics-approved field trials with blind and low-vision users.</li></ul> |                |

## TEACHING EXPERIENCE

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### Undergraduate Teaching Assistant

1+ year

Department of Computer Science and Engineering, BRAC University — Dhaka, Bangladesh

- Served as Teaching Assistant for both Data Structures and Algorithms, supporting lectures, lab sessions, and consultation hours — reaching 200+ students over a year and counting.
- Guided students through programming assignments and algorithmic problem-solving, strengthening their understanding of fundamental data structures and algorithm design.
- Graded assignments and exam scripts and gave constructive, individualized feedback to help students improve their performance.

## SELECTED PROJECTS

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### WeHeal — Healthcare Innovation Platform

[weheal.live](#)

An AI-powered, all-in-one healthcare platform for patient care and medical diagnostics — online doctor consultations, appointments, prescriptions, and emergency support in one place. Built with a team and awarded 1st Runner-Up at BRAC University's System Analysis & Design Project Showcase (2025).

### Autonomous ArUco-Marker Multi-Terrain Service Bot

[Video demo](#)

An autonomous service robot that navigates varied terrain using ArUco fiducial-marker detection for localization and task guidance, pairing computer vision with real-time control on a mobile robotic platform.

## TECHNICAL SKILLS

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**Programming Languages:** Python, Java, C++, JavaScript

**Frameworks & Libraries:** Node.js, Express.js, MediaPipe

**Databases:** MySQL, MongoDB

**Tools & Technologies:** Git, REST APIs, Machine Learning frameworks, Computer Vision libraries

## HONORS & AWARDS

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- Dean's List — recognized for academic excellence, BRAC University.
- 2nd Place — BRAC University RoboSoccer Competition.
- 1st Runner-Up — System Analysis and Design Project Showcase, Summer 2025 (for WeHeal).

## LANGUAGES

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English — Professional working proficiency · Bangla — Native proficiency

## REFERENCES

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### Dr. Aniqua Nusrat Zereen

Mahidol University — Thailand

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### Dr. Jia Uddin

Woosong University — South Korea

[engrjiauddin@gmail.com](mailto:engrjiauddin@gmail.com)